SUPPLEMENTAL DATA

Supplementary Table 1. Search strategy for each database on July 20th, 2020

Bibliographic database	Search String	Remarks
EMBASE	(``sepsis prediction'' OR ``sepsis detection'') AND (``machine learning'' OR ``artificial intelligence'')	
Google Scholar	(``sepsis prediction'' OR ``sepsis detection'') AND (``machine learning'' OR ``artificial intelligence'')	Software 'Publish or Perish' used to perform the search
Pubmed/Medline	(``sepsis prediction'' OR ``sepsis detection'') AND (``machine learning'' OR ``artificial intelligence'')	
Scopus	TITLE-ABS-KEY ((``sepsis AND prediction'' OR ``sepsis AND detection'') AND (``machine AND learning'' OR ``artificial AND intelligence''))	
Web of Science	TOPIC: ((``sepsis prediction'' OR ``sepsis detection'') AND (``machine learning'' OR ``artificial intelligence'')) Timespan: All years. Indexes: SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI, CCR-EXPANDED, IC.	

Supplementary Table 2. List of variables extracted

Category	Variable	Remark
Publication Details	Author	
	Year of Publication	
	DOI	
Demographics	Sex	count
	Age	years
	Ethnicity	count
	Number of patients with sepsis	count
	Number of controls	count
	Prevalence of sepsis	%
Study details	Dataset	
	Prediction task	
	Sepsis defintion	
	Case-control alignment	Alignment of sepsis onset in controls
	Inclusion Criteria	
Software and Data	Data of used cohort available	
	Code for analysis	
	Code for sepsis label	
	Software	
	Library versions	
	Code licence available	yes/no
Machine learning details	Model used	
	Model explanation	
	Evaluation metrics	
	External validation	
	Handling of missing data	
	Data types	
	Number of variables	
	Handling of circularity	

Supplementary Table 3. An explanation of quality assessment criteria

Categories	Items	Description	Reported
Unmet need	Limits in current non-machine-learning approach	Low diagnostic accuracy, low human-level prediction accuracy or prolonged diagnostic procedure	Yes/No
Reproducibility	Prevalence of Sepsis	The proportion of patients that sustained a sepsis out of the entire study cohort	Yes/No
	Data availability	Is the data used in the study publicly available?	Yes/No
	Feature engineering methods	How features were generated before model training	Yes/No
	Code for data wrangling and analysis	Code describing the details of the cleaning, preprocessing, and analysis of the data.	Yes/No
	Code of label	Code describing sepsis label generation	Yes/No
	Platforms/packages	Both platforms and packages should be reported	Yes/No
	Hyperparameters	All hyperparameters that are needed for study replication	Yes/No
Robustness	Sample size > 50	Sepsis case sample size > 50 is required for the interpretation, power, and validity of machine learning methods.	Yes/No
	Valid methods for over-fitting	Valid methods for unbiased performance assessment (or methods that prevent overfitting)	Yes/No
	Stability of results	Calculated variation in the validation statistics	Yes/No
Generalisability	External data validation	Validation in settings different from the research framework	Yes/No
Clinical significance	Predictors explanation	Explanation (biological or quantification) of the importance of each predictor	Yes/No
	Suggested clinical use	Clinical usability and requirements (e.g. what is still necessary before making deployment possible)	Yes/No

Supplementary Table 4. Demographic information reported by all studies (if available)

	Study	Age	Sex (% female)	Ethnicity (% non-Caucasian)
1	Abromavicius 2020	—	—	—
2	Barton 2019	55 (median, USCF), 65 (median, MIMIC-III)	54.62 (USCF), 43.87 (MIMIC-III)	—
3	Bloch 2019	55.4 (target group mean), 52.5 (control group mean)	35 (target group), 40 (control group)	—
4	Calvert 2016	—	—	—
5	Desautels 2016	65 (median)	43.85	—
6	Futoma 2017	—	—	—
7	Kaji 2019	—	—	—
8	Kam 2017	—	—	—
9	Lauritsen 2020	—	—	—
10	Lukaszewski 2008	60.16 (mean)	42.42	—
11	Mao 2018	65 (median)	43.97	—
12	McCoy 2017	—	51.43	—
13	Moor 2019	67.2 (target group mean), 64.2 (control group mean	41.4 (target group), 45.4 (control group)	27.9 (target group), 28 (control group)
14	Nemati 2018	—	52.4	—
15	Reyna 2020	—	—	—
16	Schamoni 2019	63 (median)	43.39	—
17	Scherpf 2019	—	—	—
18	Shashikumar 2017a	59 (median)	47	—
19	Shashikumar 2017b	63 (median)	48.8	—
20	Sheetrit 2019	—	38.4 (target group), 42.2 (control group)	—
21	van Wyk 2019a	—	—	—
22	van Wyk 2019b	—	—	—